

Cyber Attacks Explained

Wireless Attacks

Wireless networks are everywhere, from the home to corporate data centres. They make our lives easier by avoiding bulky cables and related problems. But with these benefits comes a threat: wireless networks are prone to attacks. This article discusses techniques to protect FOSS networks, which systems administrators can implement to achieve adequate security.



Before we talk about wireless security and vulnerability attacks, we must understand the basic radio transmissions, and the IEEE 802.11 protocol, also commonly known as the WLAN protocol. This protocol links two or more devices over a short distance, using spread spectrum signals. Spread spectrum, at its core, is based on radio communication frequencies to establish point-to-point wireless communication between a transmitter and a receiver, while achieving resistance to signal jamming and signal fading. As shown in Figure 1, to establish a wireless network, you need a wireless access point (AP) and also a wireless adaptor for each node to be connected. The AP is also called a hot-spot; it hosts a radio transceiver similar to a walkie-talkie. It also contains hardware to convert digital data into radio signals and vice-versa.

The AP has a unique feature called a beacon transmission, whereby it keeps transmitting a digitised signal, typically, a few times every second. This signal contains the network identification data, the service set identifier (SSID) and some trivial error-correction information. Nodes such as laptops or other wireless devices detect this signal in order to show it in the list of available wireless networks. It also detects whether or not the AP is using any security, the level of the security protocol, etc.

The AP contains a TCP/IP stack, which responds to ARP requests when a node tries to connect to it. Since wireless

networks can allow multiple nodes, it is essential to have an authentication layer prior to letting data transfer take place. It is the AP's responsibility to ensure this security, as well as to monitor packet transmission and data integrity.

Wi-Fi security

Since wireless networks don't have built-in security mechanisms, a secure layer on top of the wireless protocol stack is achieved by encryption and authentication techniques such as WEP (Wired Equivalent Privacy) or WPA (Wi-Fi Protected Access). This is especially important because, unlike a wired network, wireless signals can be easily intercepted using a signal-trapping device. Let's discuss how these encryption techniques work, in detail.

To establish a secure channel, the client first sends an authentication request to the AP, and receives a challenge from it in text form. The client encrypts this text using the preconfigured key and sends it back. The AP decrypts it, and when it succeeds, replies to the client. If the keys don't match, the request is dropped, and the client cannot connect to the AP. This method is called pre-shared authentication. In an improved version, the shared key is combined with the SSID of the wireless AP, to further toughen the encryption key logic. WEP encryption uses the RC4 algorithm on all packets that travel between the AP and the node.